THE UNIVERSITY OF HONG KONG
FACULTY OF BUSINESS AND ECONOMICS

MAIB7014 Human-AI Interaction and Collaboration

GENERAL INFORMATION

Instructor: Prof. Zhenhui (Jack) Jiang

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Consultation times: By appointment

Tutor: TBD

Pre-requisites: [ ]
Co-requisites: [None]
Mutually exclusive: N/A

Course Website: TBD
Reference Book:
[Book]
The following programming languages are used in this course: [programing language]
[Software] is also used.

COURSE DESCRIPTION

Artificial Intelligence (AI) is increasingly embedded in the ways people work, learn, communicate, and make decisions. As a result, the success of AI systems depends not only on technical performance, but also on how effectively humans understand, use, supervise, and collaborate with them. This course introduces the foundations of Human–AI Interaction (HAI), with a focus on designing and evaluating AI systems that are useful, reliable, trustworthy, safe, and aligned with human needs and values.

This course explores the dynamic interplay between humans and artificial intelligence systems. Throughout the course, students will engage with a blend of theoretical foundations and practical applications, gaining insights into the cognitive, social, technical, and ethical dimensions of human-AI interaction and collaboration. Key topics include user-centred design for AI, human factors in AI systems, interactive machine learning, explainable AI, human-robots interaction and collaboration, and the role of AI in augmenting human capabilities.

Through lectures, discussions, case analyses, and hands-on project work, students will learn how to analyse user needs, design AI-enabled interactions, prototype solutions, and evaluate them using human-centred methods. By the end of the course, students will be able to think critically about the role of AI in society and design Human–AI systems that support meaningful human control and responsible innovation.

COURSE OBJECTIVES

The course aims to help students:

1. develop a foundational understanding of key concepts, theories, and design principles in Human–AI Interaction;
2. examine how humans interact with, interpret, and rely on AI systems in different contexts;

3. understand the major design, ethical, and governance challenges associated with AI, including trust, explainability, fairness, privacy, hallucination, and accountability;
4. gain practical experience in analysing, designing, and evaluating human-centred AI systems;
5. cultivate critical and reflective thinking about the opportunities, limitations, and societal implications of AI.

PROGRAMME LEARNING OUTCOMES

PLO1: Acquisition of the techniques and advanced knowledge of the programme discipline

PLO2: Application and integration of knowledge

PLO3: Inculcating professionalism and leadership

PLO4: Developing global outlook

PLO5: Mastering communication skills

COURSE LEARNING OUTCOMES

Course Learning Outcomes	Aligned Programme Learning Outcomes
CLO1: [Explain major concepts, theories, and design principles in Human–AI Interaction]	PLO[1, 5]
CLO2: [Analyse how humans perceive, interpret, trust, and collaborate with AI systems in different contexts]	PLO[1, 2 , 3]
CLO3: [Apply human-centred design methods to investigate user needs and develop AI-enabled solutions for real-world problems]	PLO[1, 2, 3, 4]
CLO4: [Evaluate Human–AI interaction designs using appropriate user research methods, metrics, and critical reflection on system limitations and societal impact]	PLO[1, 2, 3, 4, 5]

COURSE TEACHING AND LEARNING ACTIVITIES

Course Teaching and Learning Activities	Expected contact hour	Study Load (% of study)
T&L1. Interactive lectures	30	25%
T&L2. Group projects discussion	30	25%
T&L3. Assignments / Tutorials	20	16.7%
T&L4. Lecture preparation & Self-study	40	33.3%
Total	120	100%

Assessment Methods	Brief Description (Optional)	Weight	Aligned Course Learning Outcomes
A1. Class participation	Attendance & discussions	30%	CLO1, 2, 4
A2. Quizzes and Assignments	Effort and accuracy	30%	CLO1, 2, 3, 4
A3. Project	Effort and accuracy	40%	CLO1, 2, 4
	Total	100%	

STANDARDS FOR ASSESSMENT

Course Grade Descriptors

A+, A, A-	• Demonstrate a strong understanding of all relevant knowledge • Handling questions professionally • High participation in discussions • Present arguments that have an element of originality

	・Achieve a standard of excellent performance in the exams with very accurate computation and very good analytical and problem solving skills ・Excellent performance in assignments
B+, B, B-	・Demonstrate a good understanding of all relevant knowledge ・Handling questions in a logical way ・Good participation in discussions ・Present arguments that go beyond the lecture and textbook ・Achieve a standard of good performance in the exams with accurate computation and good analytical and problem solving skills ・Good performance in assignments
C+, C, C-	・Demonstrate a basic understanding of the concepts involved ・Fairly address questions as set ・Some participation in discussions ・Present arguments in a well-structure manner ・Meet a standard of acceptable performance in the exams with reasonably accurate computation and acceptable analytical and problem solving skills ・Acceptable performance in assignments
D+, D	・Demonstrate a minimum understanding of the concepts involved ・Barely address questions as set ・Minimal or no participation in discussions ・Present arguments in a marginally acceptable manner ・Meet a standard of marginally acceptable performance in the exams with some errors in computation and barely adequate analytical and problem solving skills ・Marginally acceptable performance in assignments
F	・Demonstrate a poor understanding of the concepts involved ・Unable or unwilling to handle questions ・Minimal or no participation in discussions ・Present arguments poorly ・Fail to meet a standard of passing the exams with major errors in computation and inadequate analytical and problem solving skills ・Poorly performance in assignments
Assessment Rubrics for Participation	
A+, A, A-	・High participation in discussions ・Always attend in-class discussions ・Demonstrate a strong understanding of all relevant knowledge ・Handling questions professionally ・Present arguments that have an element of originality ・Respect others and follow the class rules (no chatting and do not use cell phone)
B+, B, B-	・Good participation in discussions ・Often attend in-class discussions ・Demonstrate a good understanding of all relevant knowledge ・Handling questions in a logical way ・Present arguments that go beyond the lecture and textbook ・Respect others and follow the class rules (no chatting and do not use cell phone)
C+, C, C-	・Some participation in discussions ・Sometimes attend in-class discussions ・Demonstrate a basic understanding of the concepts involved ・Fairly address questions as set ・Present arguments in a well-structure manner ・Respect others and follow the class rules (no chatting and do not use cell phone)
D+, D	・Minimal or no participation in discussions ・Rarely attend in-class discussions ・Demonstrate a minimum understanding of the concepts involved ・Barely address questions as set ・Present arguments in a marginally acceptable manner ・Respect others and follow the class rules (no chatting and do not use cell phone)
F	・Minimal or no participation in discussions ・Almost never attend in-class discussions

	• Demonstrate a poor understanding of the concepts involved • Unable or unwilling to handle questions • Present arguments poorly • Behave poorly in class (often chatting with others, using cell phones, or being late)
Assessment Rubrics for Assignments, Quizzes, and Projects	
A+, A, A-	• Demonstrate a strong understanding of all relevant knowledge • Present arguments that have an element of originality • Achieve a standard of excellent performance in the assessments with very accurate computation and very good analytical and problem solving skills
B+, B, B-	• Demonstrate a good understanding of all relevant knowledge • Present arguments that go beyond the lecture and textbook • Achieve a standard of good performance in the assessments with accurate computation and good analytical and problem solving skills
C+, C, C-	• Demonstrate a basic understanding of the concepts involved • Present arguments in a well-structure manner • Meet a standard of acceptable performance in the assessments with reasonably accurate computation and acceptable analytical and problem solving skills
D+, D	• Demonstrate a minimum understanding of the concepts involved • Present arguments in a marginally acceptable manner • Meet a standard of marginally acceptable performance in the assessments with some errors in computation and barely adequate analytical and problem solving skills
F	• Demonstrate a poor understanding of the concepts involved • Present arguments poorly • Fail to meet a standard of passing the assessments with major errors in computation and inadequate analytical and problem solving skills

COURSE CONTENT AND TENTATIVE TEACHING SCHEDULE

Session	Topics
1	Introduction to Human-Centered AI
2	Human Agency, Control, and Automation
3	AI Errors, Hallucination, and Trust Calibration
4	Fairness, Bias, and Inclusive AI
5	Explainability and Transparency
6	Privacy, Security, and Responsible Data Use
7	Human-AI Collaboration and Decision Support Industrial Guest Talk
8	Evaluating Human-AI Systems
9	Governance, Accountability, and Societal Impact
10	Future Directions in Human-Centered AI

REQUIRED/RECOMMENDED READINGS & ONLINE MATERIALS (e.g. journals, textbooks, website addresses etc.)

To be announced on Moodle.

MEANS/PROCESSES FOR STUDENT FEEDBACK ON COURSE

X conducting mid-term survey in additional to SFTL around the end of the semester

○ Online response via Moodle site

○ Others: _____ (please specify)

COURSE POLICY (e.g. plagiarism, academic honesty, attendance, etc.)

An orderly learning environment is extremely important for this course. Disruptive behaviors are absolutely unacceptable. Academic dishonesty includes cheating, plagiarism, unauthorized collaboration, falsifying academic records, and any act designed to avoid participating honestly in the learning process. Any such dishonesty will result in an F grade.

ADDITIONAL COURSE INFORMATION (e.g. e-learning platforms & materials, penalty for late assignments, etc.)

Assignments will be submitted in class before the lectures. No late assignment will be accepted.
Lecture notes and self-learning materials will be uploaded on Moodle.
Good questions and discussions on Moodle will be awarded participation grades.